SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. *(BL3)*

Assessment Tool Weightage: 40%

Duration :60 Mins
Off Class
QUESTIONS: 5 **Total Marks:50**

Annexure A

DESIGN TASK

Code of Conduct:

I ARUNPRASATH R (192524077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
ARUNPRASATH R

Annexure A

DESIGN TASK

<table>
<tr><th>S.No</th><th>Register Number</th><th>Questions</th></tr>
<tr><td rowspan="2">3</td><td rowspan="2">192524077</td><td>Design a parser to recognize Boolean expressions.
Grammar
B → B && B
B → B || B
B → !B
B → (B)
B → true
B → false
Input
true && false || true
Design Requirements
(i) Implement the lexical analyzer using Lex.
(ii) Implement the parser.
(iii) Validate the Boolean expression.
(iv) Display Accepted or Rejected.
(v) Handle syntax errors.</td></tr>
<tr><td>Design a parser to recognize variable declarations.
Grammar
D → Type IdList ;
Type → int | float | char
IdList → id
IdList → id , IdList
Input
int a, b, c;
Design Requirements
(i) Implement the lexical analyzer using Lex.
(ii) Implement the parser.
(iii) Validate the declaration.
(iv) Display Accepted or Rejected.</td></tr>
</table>

(v) Handle syntax errors.

Design a parser to recognize assignment statements.

Grammar

S → id = E ;

E → E + T

E → T

T → id

T → num

Input

x = a + 10;

Design Requirements

(i) Implement the lexical analyzer using Lex.

(ii) Implement the parser.

(iii) Validate the assignment statement.

(iv) Display Accepted or Rejected.

(v) Handle syntax errors.

Design a parser to recognize function declarations.

Grammar

F → Type id ( ) Block

Type → int | float | void

Block → { }

Input

int display()

{

}

Design Requirements

(i) Implement the lexical analyzer using Lex.

(ii) Implement the parser.

(iii) Validate the function declaration.

(iv) Display Accepted or Rejected.

(v) Handle syntax errors.

Design a parser to recognize function calls.

Grammar

Call → id (ArgList)

ArgList → id

		ArgList → id , ArgList ArgList → ε **Input** sum(a,b,c) **Design Requirements** (i) Implement the lexical analyzer using Lex. (ii) Implement the parser. (iii) Validate the function call. (iv) Display Accepted or Rejected. (v) Handle syntax errors.

1.

```
C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex --version
flex version 2.5.4

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>dir C:\GnuWin32\bin
 Volume in drive C is OS
 Volume Serial Number is BC6A-BF46

 Directory of C:\GnuWin32\bin

16-07-2026  09:07    <DIR>          .
16-07-2026  09:08    <DIR>          ..
05-05-2009  01:47           279,552 bison.exe
04-09-2022  18:36                54 bison.yacc
08-04-2004  03:26           170,496 flex++.exe
08-04-2004  03:26           170,496 flex.exe
15-03-2008  03:51         1,008,128 libiconv2.dll
07-05-2005  01:22           103,424 libintl3.dll
03-04-2009  04:45           179,200 m4.exe
24-10-2007  15:40            79,360 regex2.dll
27-12-2010  22:40            77,824 sed.exe
04-09-2022  18:36                54 yacc
              10 File(s)      2,068,588 bytes
               2 Dir(s)  326,543,646,720 bytes free

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex boolean.l
flex: can't open boolean.l

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>dir
 Volume in drive C is OS
 Volume Serial Number is BC6A-BF46

 Directory of C:\Users\arunp\OneDrive\Documents\Compiler - LAB

05-08-2026  16:41    <DIR>          .
29-07-2026  12:14    <DIR>          ..
30-07-2026  11:54            49,631 a.exe
05-08-2026  16:41               340 boolean.l.txt
05-08-2026  16:41               547 boolean.y.txt
29-07-2026  11:56               232 exp25.l.txt
29-07-2026  12:15               317 exp26.l.txt
29-07-2026  12:19               238 exp27.l.txt
29-07-2026  12:33               889 exp29.l.txt
30-07-2026  11:21            49,679 exp30.exe
30-07-2026  11:20               453 exp30.l.txt
30-07-2026  11:32               337 exp31.l.txt
30-07-2026  11:37               399 exp32.l.txt
30-07-2026  11:46               194 exp33.l.txt
30-07-2026  11:54               298 exp34.l.txt
28-07-2026  14:41               253 exp35.l.txt
28-07-2026  14:46               280 exp37.l.txt
28-07-2026  14:57               238 exp39.l.txt
28-07-2026  14:32               389 exp4.l.txt
29-07-2026  12:01            49,631 exp40.exe
28-07-2026  14:24               313 ignore.l.txt
30-07-2026  11:54            37,692 lex.yy.c
29-07-2026  12:35                79 sample.c
              21 File(s)        192,429 bytes
               2 Dir(s)  326,466,473,984 bytes free

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex boolean.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>dir lex.yy.c
 Volume in drive C is OS
 Volume Serial Number is BC6A-BF46
```

```
              2 Dir(s)  324,382,322,112 bytes free

:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\MinGW\bin;C:\GnuWin32\bin;%PATH%

:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc --version
gcc (MinGW.org GCC-6.3.0-1) 6.3.0
opyright (C) 2016 Free Software Foundation, Inc.
his is free software; see the source for copying conditions.  There is NO
arranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.


:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o boolean.exe

:\Users\arunp\OneDrive\Documents\Compiler - LAB>boolean.exe
nter Boolean Expression:
rue && false || true
^Z

ccepted

:\Users\arunp\OneDrive\Documents\Compiler - LAB>
```

2.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex decl.l.txt
'flex' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:\MinGW\bin;%
PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex --version
flex version 2.5.4

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc --version
gcc (MinGW.org GCC-6.3.0-1) 6.3.0
Copyright (C) 2016 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.


C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex decl.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy decl.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o decl.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>decl.exe
Enter Variable Declaration:
^C
C:\Users\arunp\OneDrive\Documents\Compiler - LAB>decl.exe
Enter Variable Declaration:
int a,b,c;

Accepted
```

3.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:
\MinGW\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex assign.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy assign.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o ass
ign.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>assign.exe
Enter Assignment Statement:
x = a + 10;

Accepted
```

4.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:
\MinGW\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex funcdecl.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy funcdecl.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o fun
cdecl.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>funcdecl.exe
'C:\Users\arunp\OneDrive\Documents\Compiler - LAB\funcdecl.exe' was blocked
by your organization's Device Guard policy.
Contact your support person for more info.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o a.e
xe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>a.exe
Enter Function Declaration:
int display()
{
}

Accepted
```

5.

```
C:\Windows\System32\cmd.e
(c) Microsoft Corporation. All rights reserved.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>set PATH=C:\GnuWin32\bin;C:\MinGW\bin;%PATH%

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>flex funccall.l.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>bison -dy funccall.y.txt

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o funccall.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>funccall.exe
'C:\Users\arunp\OneDrive\Documents\Compiler - LAB\funccall.exe' was blocked by your organization's Device Guard policy.
Contact your support person for more info.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>a.exe
Enter Function Declaration:
sum(a,b,c)

Rejected

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o a.exe
c:/mingw/bin/../lib/gcc/mingw32/6.3.0/../../../../mingw32/bin/ld.exe: cannot open output file a.exe: Permission denied
collect2.exe: error: ld returned 1 exit status

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>taskkill /F /IM a.exe
SUCCESS: The process "a.exe" with PID 896 has been terminated.

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>gcc lex.yy.c y.tab.c -o a.exe

C:\Users\arunp\OneDrive\Documents\Compiler - LAB>a.exe
Enter Function Call:
sum(a,b,c)

Accepted
```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and integrates multiple concepts effectively	Good understanding with proper integration of most concepts	Basic understanding with partial integration	Poor understanding with no integration
Design & Implementation	30	Well-designed solution with systematic and optimal	Correct design with minor issues	Basic design with errors or inefficiencies	Incorrect or incomplete design

		implementation			
Parser/Algorithm Construction	20	Accurate construction of parser/algorithm with complete logic	Mostly correct construction with minor errors	Partial construction with missing logic	Incorrect or incomplete construction
Analysis & Validation	15	Insightful analysis with correct validation and justification	Good analysis with reasonable justification	Basic analysis with limited validation	Poor or incorrect analysis
Documentation & Presentation	10	Well-structured documentation with diagrams, steps, and clarity	Organized documentation with required details	Basic documentation with missing details	Poor or incomplete documentation